

```

lats.append(lat)

print(mags[:10])
print(lons[:5])
print(lats[:5])

```

We make empty lists for the longitudes and latitudes. The code `eq_dict['geometry']` accesses the dictionary representing the geometry element of the earthquake ❶. The second key, 'coordinates', pulls the list of values associated with 'coordinates'. Finally, the 0 index asks for the first value in the list of coordinates, which corresponds to an earthquake's longitude.

When we print the first 5 longitudes and latitudes, the output shows that we're pulling the correct data:

```

[1.6, 1.6, 2.2, 3.7, 2.92000008, 1.4, 4.6, 4.5, 1.9, 1.8]
[-150.7585, -153.4716, -148.7531, -159.6267, -155.248336791992]
[61.7591, 59.3152, 63.1633, 54.5612, 18.7551670074463]

```

With this data, we can move on to mapping each earthquake.

Building a World Map

Using the information we've pulled so far, we can build a simple world map. Although it won't look presentable yet, we want to make sure the information is displayed correctly before focusing on style and presentation issues. Here's the initial map:

```

eq_world
_map.py  from pathlib import Path
import json

import plotly.express as px

--snip--
for eq_dict in all_eq_dicts:
    --snip--

title = 'Global Earthquakes'
❶ fig = px.scatter_geo(lat=lats, lon=lons, title=title)
fig.show()

```

We import `plotly.express` with the alias `px`, just as we did in Chapter 15. The `scatter_geo()` function ❶ allows you to overlay a scatterplot of geographic data on a map. In the simplest use of this chart type, you only need to provide a list of latitudes and a list of longitudes. We pass the list `lats` to the `lat` argument, and `lons` to the `lon` argument.

When you run this file, you should see a map that looks like the one in Figure 16-7. This again shows the power of the Plotly Express library; in just three lines of code, we have a map of global earthquake activity.